

IMTOP, IMage-TO-Print wound dressings: from a single calibrated RGB photograph to a patient-specific patch

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Abstract

Background and objective: Personalized diabetic foot ulcer (DFU) dressings must follow the individual lesion, yet existing image-to-fabrication pipelines depend on depth sensors or three-dimensional reconstruction and validate the segmentation mask rather than the fabricated object. We present a low-cost pipeline that turns a single calibrated RGB photograph into a printable patient-specific patch, with a validation framework centred on the patch itself.

Methods: Absolute scale is recovered from two operator-placed points on a reference of known length, without camera intrinsics or depth. The wound contour is fitted with a closed degree-five B-spline, offset by a perilesional margin and extruded to a watertight STL solid. The fabricated-patch error is decomposed into segmentation (E_{Seg}), scale (E_{Scale}) and printing (E_{Pr}) terms, each measured against a known reference on a dedicated experiment. The pipeline was evaluated on 149 clinical DFU images from two public datasets (single-operator reference, three segmentation back-ends), 18 fused-deposition-modelled phantoms of known CAD geometry and a Monte-Carlo calibration analysis.

Results: Segmentation was the largest error term by magnitude: for the default back-end (Segment Anything Model, ViT-B) E_{Seg} was 6.4%, against 5.0% for scale and 0.69% for printing, for a combined area error of 8.1% with printing negligible throughout. The three back-ends produced similar region overlap (Dice 0.950, 0.953 and 0.942 for SAM, GrabCut and the

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marker-controlled watershed), which nonetheless differed overall (Friedman $\chi^2 = 24.98$, $p < 0.001$), the only significant pairwise contrast being GrabCut over watershed; watershed failed on 3 of the 149 images. On the phantoms the in-plane geometry was reconstructed to 0.32% area error (Dice 0.997), locating the clinical error in the wound-border ambiguity rather than in scale recovery or the algorithm. Mask overlap was correlated with the patch error but did not determine it: at Dice ≥ 0.95 the patch area error still spanned 0 to 8.9%, so overlap was necessary but insufficient, and the residual under-coverage was converted into a back-end-specific perilesional offset.

Conclusions: A single calibrated RGB photograph suffices to produce a patient-specific DFU dressing without depth sensing. Segmentation is the largest geometric error term by magnitude and printing is negligible, region overlap alone does not fully determine patch accuracy, and the residual segmentation uncertainty yields a constructive perilesional-offset rule. The application will be released to support reproducibility.

Keywords: Diabetic foot ulcer, Wound dressing, Image segmentation, 3D printing, Error propagation, Coverage analysis

1. Introduction

Chronic skin wounds, and diabetic foot ulcers (DFUs) in particular, are a growing clinical and economic burden [1]. The lifetime incidence of a foot ulcer in people with diabetes is high, and its consequences are severe, including lower-limb amputation and a post-ulceration mortality comparable to that of several common cancers [2, 3]. A recurring cause of poor healing is non-personalized treatment: standardized protocols and pre-shaped dressings ignore wound-specific features such as contour, area and local tissue condition [1], and pre-shaped dressings conform poorly to irregular morphologies even though close contact with the wound bed is considered important for efficacy [4].

This has motivated personalized wound care. Hydrogel dressings are widely used because they keep the wound moist, a condition that accelerates epithelialization [5], while absorbing exudate, allowing gas exchange and releasing therapeutics locally [6]. To match the dressing to the individual lesion, additive manufacturing has been used to fabricate patient-specific shapes, from three-dimensionally scanned antimicrobial patches to printed functional hydrogels [4, 7, 8]. The geometric step, turning a clinical image

of the wound into a fabricable object, has so far relied mainly on RGB-D pipelines that acquire depth and reconstruct a three-dimensional model before generating printable instructions [9]. Two limitations follow. Such pipelines raise the hardware requirement and are ill-suited to point-of-care acquisition, where only a consumer camera may be available [9]; and the geometric result is usually validated on the segmentation mask with region-overlap metrics, not on the object that is finally fabricated [10, 11].

Diabetic foot ulcers range from superficial partial- or full-thickness lesions to deep wounds that reach tendon or bone [12]. This work targets the superficial class treated with conformal dressings, for which a patch that reproduces the wound contour and carries a parametric thickness is a reasonable design choice; deep or undermined wounds fall outside its scope. For this class a single RGB photograph is geometrically sufficient: absolute scale is recovered from a physical reference in the field of view, as in image-based wound planimetry [13], with no depth sensor and no reconstruction of the wound-bed topography. The contribution of such a low-cost, depth-free pipeline is not the patch itself, which is a simple geometric transformation, but the characterization of how the upstream errors reach the fabricated object and the conversion of that characterization into a design rule. Manufacturing routinely decomposes the deviation of a produced part across successive process stages and recombines it, an error-budget perspective that, to our knowledge, has not been applied to image-to-print wound dressings [14, 15].

We organize the study around three research questions:

- RQ1 How does the fabricated-patch error distribute across the segmentation, scale and printing stages, in both magnitude and variance?
- RQ2 Is mask overlap a sufficient predictor of fabricated-patch accuracy, or are boundary and coverage measures also required?
- RQ3 Can the residual segmentation uncertainty be turned into a perilesional offset that covers the wound at a target level?

We address them through three contributions:

- a stage-wise error budget that splits the fabricated-patch error into segmentation (E_{Seg}), scale (E_{Scale}) and printing (E_{Pr}) terms, each measured against a known reference on a dedicated experiment (RQ1);

- a fabrication-aware evaluation that complements region overlap with boundary metrics and with clinically asymmetric coverage metrics, separating unsafe under-coverage of the wound from lower-cost over-coverage onto healthy skin (RQ2);
- an uncertainty-driven sizing of the perilesional offset, yielding the margin that covers the wound at a target level, on the evaluated data, for a chosen segmentation back-end (RQ3).

The pipeline, including the interactive application used to produce the manual reference segmentations and the patches, will be released to support reproducibility (Code availability). The rest of the paper presents related work (Section 2), the system and methods (Section 3), the results (Section 4), a discussion with the study limitations (Section 5) and the conclusions (Section 6).

2. Related work

Personalized wound care has produced many engineered dressings, and additive manufacturing gives them a patient-specific shape: three-dimensionally scanned antimicrobial polycaprolactone patches [4], pH-indicating auxetic hydrogel dressings [7] and bioprinted constructs [8]. Hydrogels are preferred because they keep the wound moist, which accelerates epithelialization [5], while absorbing exudate and releasing drugs locally [6]. Since conformal contact with the wound bed matters for efficacy [4], the geometric fidelity of the patch is itself a design target.

Extracting geometry from a clinical image follows two routes. The first stays on plain RGB: planimetry recovers wound area from a photograph with a physical scale reference [13], mobile apps standardize acquisition on a smartphone [16], deep models segment wound tissue on-device [17], and phone photographs predict ulcer healing [18]; these report area, tissue or prognosis but not a fabricable object. The second reconstructs three-dimensional geometry: robot-mounted RGB-D rigs reconstruct the wound surface [19] and derive measurements [20], Cavazzana et al. add a three-dimensional analysis for objective assessment [21], and Chae et al. reconstruct an RGB-D surface to print bioprintable DFU patches [9]. This route needs a depth sensor or a controlled rig, ill-suited to point-of-care capture, and, except for [9], validates a measurement or surface rather than a fabricated object. IMTOP keeps the

depth-free acquisition of the first route while reaching the patch, and follows an assessment pipeline such as [21].

The segmentation feeding these pipelines has a large literature. Classical region methods, Otsu thresholding [22], marker-controlled watershed [23] and GrabCut [24], were superseded by deep learning: fully convolutional networks [25], U-Net [26], applied to DFUs by Goyal et al. [11], to general wounds by Wang et al. [10] and through CNN ensembles [27], with self-configuring nnU-Net [28]; public challenges rank clinician-delineated masks by overlap [29]. Promptable foundation models, the Segment Anything Model [30] and its medical adaptation MedSAM [31], segment from a box or point prompt without task-specific training. In almost all of this work the result is scored on the mask; IMTOP reuses three such back-ends only as interchangeable front-ends to its geometric chain.

That scoring almost always uses region overlap, the Dice coefficient [32] and the intersection over union. Overlap is an incomplete descriptor: two masks with identical Dice can differ in boundary placement, captured by the Hausdorff distance [33], and it ignores the clinical asymmetry between exposing wound and extending onto healthy skin. Both limitations become concrete once the mask becomes a physical patch, which motivates boundary and coverage measures and measuring the error on the fabricated object.

On the fabrication side, the deviation of a produced part is routinely split across process stages and recombined into an error budget, following the guide to uncertainty propagation in measurement [14] and experimental studies of dimensional accuracy in fused-deposition modelling [15, 34]; similar metrological protocols address fabricated medical devices such as surgical guides [35]. To our knowledge this perspective has not been applied to image-to-print wound dressings, where the stages are segmentation, scale recovery and printing.

Table 1 places IMTOP among these methods. It propagates the segmentation, scale and printing errors to the fabricated patch, complements overlap with boundary and coverage metrics, and turns the residual segmentation uncertainty into a perilesional offset, thus extending mask-only segmentation studies [11, 10, 29] and depth-based reconstruction pipelines [9, 19, 20] toward a depth-free, fabrication-aware route validated on the produced object.

Table 1: Methods and pipelines relevant to image-based wound geometry, grouped from general-purpose segmentation to fabrication. Only IMTOP reaches a printable patch from a single RGB photograph, without depth, and reports the accuracy of the fabricated object. In the Depth column a filled circle (●) marks methods that require a depth sensor or a 3D rig, an open circle (○) image-only methods.

Work	Year	Depth	Output	Validated on
<i>General-purpose segmentation</i>				
Long et al. [25]	2015	○	semantic	overlap
Ronneberger et al. [26]	2015	○	mask	overlap
Isensee et al. [28]	2021	○	mask	overlap
Kirillov et al. [30]	2023	○	promptable	overlap
Ma et al. [31]	2024	○	promptable	overlap
<i>Wound and DFU segmentation</i>				
Goyal et al. [11]	2017	○	mask	overlap
Wang et al. [10]	2020	○	mask	overlap
Mahbod et al. [27]	2022	○	mask	overlap
Ramachandram et al. [17]	2022	○	tissue	overlap
Yap et al. [29]	2024	○	mask	overlap
<i>Planimetry, mobile and monitoring</i>				
Yap et al. [16]	2018	○	std. image	repeatability
Foltynski [13]	2018	○	area	clinical area
Kim et al. [18]	2020	○	healing risk	clinical outcome
<i>3D reconstruction and fabrication</i>				
Filko et al. [19]	2021	●	3D surface	reconstruction
Chae et al. [9]	2022	●	patch (print)	3D surface
Filko & Nyarko [20]	2023	●	mask + 3D	measurement
Cavazzana et al. [21]	2025	●	mask + 3D	assessment
IMTOP	–	○	patch (print)	fabricated patch

3. Materials and methods

3.1. System and software architecture

IMTOP is a desktop application built around a Python back-end and an HTML and JavaScript front-end embedded in a Qt WebEngine view, the two sides communicating through a Qt WebChannel bridge (Fig. 1). The front-end provides an interactive image canvas, a two-point scale tool, a segmenter selector, a parameter panel, a per-metric overlay list and a three-dimensional patch viewer; the patch mesh is built and rendered on the client with Three.js, and the STL file is written on the client from the displayed mesh. The back-end performs image input and output, scale computation, segmentation

with three interchangeable back-ends together with mask post-processing, and metric computation, and it handles all file export. The workflow is organized as a six-stage pipeline (Fig. 2): image loading, two-point scale calibration, manual reference delineation, automatic segmentation, computation of mask-level, patch-level and coverage metrics, and parametric construction of a printable patch. The pipeline is governed by a state machine whose stages are unlocked by specific back-end events, the automatic-segmentation event unlocking both the metrics and the patch stages; backward navigation is always available and loading a new image resets the pipeline. The mapping from this conceptual pipeline to the implemented interface is shown in Fig. 3.

Communication between the two sides is asynchronous: the front-end invokes back-end slots (for example `loadImage`, `setScale`, `runAutoSegmentation`, `getContours`) and the back-end emits signals (for example `imageReady`, `scaleSet`, `autoSegDone`, `metricOverlayReady`) that update the interface. The segmentation model is loaded lazily, only on its first use, on a worker thread that reads the checkpoint from disk in chunks and reports byte-level progress, so the interface never blocks; the weights are read once and reused, and only the per-image encoding pass is repeated for a new image. The application will be released to support reproducibility (Code availability).

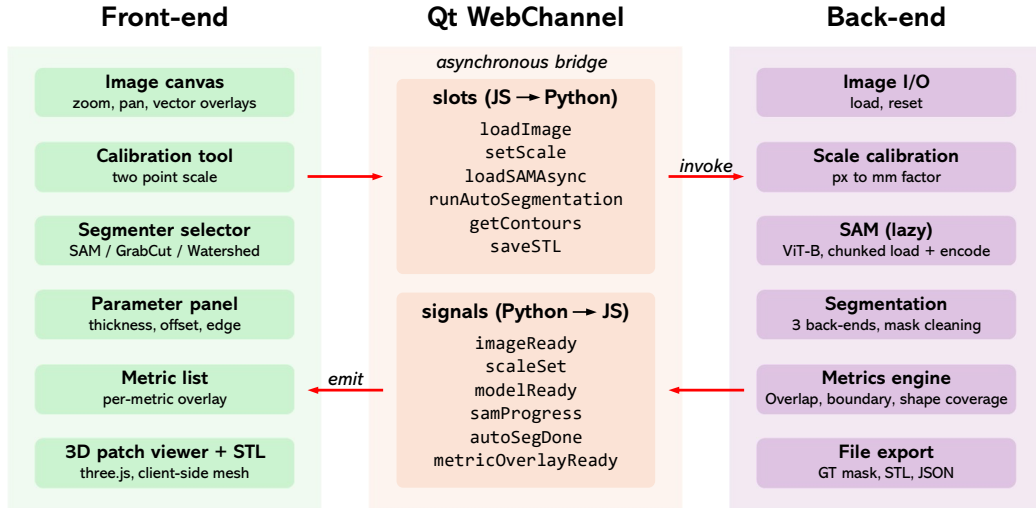


Figure 1: Software architecture: an HTML/JavaScript front-end and a Python back-end over a Qt WebChannel bridge, with the implemented slots and signals on the bus.

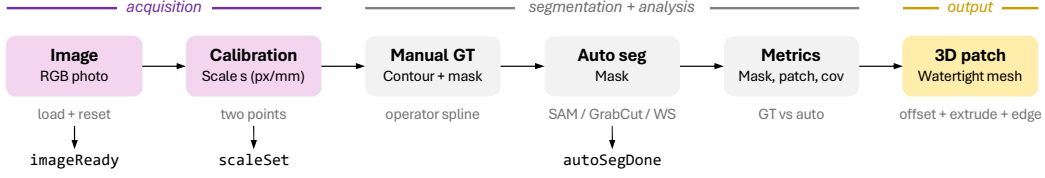


Figure 2: Six-stage pipeline from a single RGB photograph to a printable dressing; the automatic-segmentation event unlocks the metric and patch stages.

3.2. Metric scale calibration from a physical reference

Absolute scale is recovered from two points placed by the operator on a reference of known length d visible in the photograph, such as a caliper jaw opening or a ruler segment. Given the pixel coordinates $\mathbf{p}_0$ and $\mathbf{p}_1$ of the endpoints, the pixel-to-millimetre scale factor is

$$s = \frac{\|\mathbf{p}_1 - \mathbf{p}_0\|_2}{d} \quad [\text{px/mm}], \quad (1)$$

and every downstream geometric quantity is expressed in millimetres through s . Because s is the only source of absolute dimension, its uncertainty is isolated and characterized separately (Section 4.3). The formulation needs neither camera intrinsics nor a depth sensor, which is the basis of the device-agnostic property of the method; its single assumption, that the reference and the wound lie on an approximately fronto-parallel plane, is discussed in Section 5.

3.3. From contour to printable patch

The mask is converted to a printable patch through a deterministic geometric chain (Algorithm 1). The largest external contour is extracted and fitted with a closed periodic B-spline of degree five resampled to $N = 300$ points, which removes the pixel staircasing of the raster boundary while preserving the overall shape. The contour is converted to millimetres through s and offset outward, along the radial direction from its centroid, by a perilesional margin δ that extends the patch over the surrounding healthy skin; δ is not fixed a priori but derived from the coverage analysis of Section 3.6. The offset contour is extruded to a user-set thickness t (3 mm by default) with a flat top, and the bottom rim is optionally finished with a 45° chamfer or a circular fillet of radius r sampled over $N_{\text{arc}} = 20$ rings, with r clamped to the thickness and $r = 0$ leaving a square edge. Because the caps and

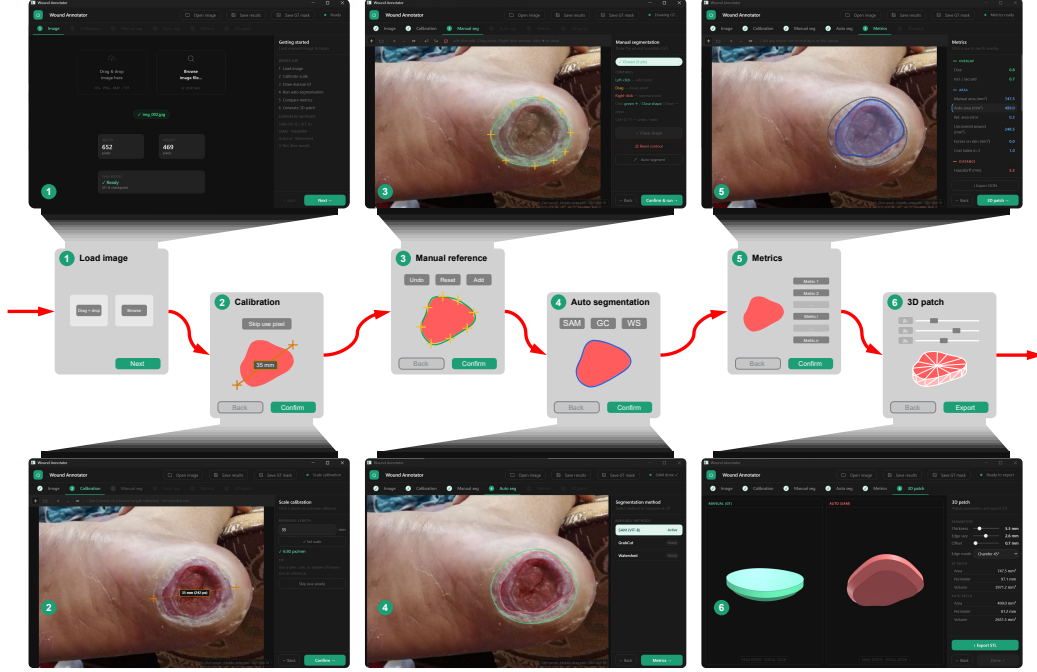


Figure 3: From pipeline concept to implemented interface. The central row sketches each stage as an interface mock-up, and the screenshots above and below show the corresponding view in the released application.

the lateral wall share their boundary vertices, the mesh is watertight and STL-exportable, and its volume follows from the divergence theorem,

$$V = \frac{1}{6} \left| \sum_{f \in \mathcal{F}} \mathbf{v}_1^{(f)} \cdot (\mathbf{v}_2^{(f)} \times \mathbf{v}_3^{(f)}) \right|, \quad (2)$$

with $\mathcal{F}$ the oriented triangular faces and $\mathbf{v}_{1,2,3}^{(f)}$ the vertices of face f . The patch is therefore a planar solid of parametric thickness with a finished edge: conformity is delegated to the flexibility of the thin dressing rather than to a built-in curvature, and the wound-bed topography is not reconstructed, a scope limit stated in Section 5. For the manual reference, steps 1 and 2 of Algorithm 1 are replaced by a closed cubic interpolating spline through the operator control points (Section 3.4); the remaining steps are identical, so the two patches are built by the same operator and are directly comparable.

Algorithm 1: Mask-to-patch geometric chain

Data: binary mask M , scale s , thickness t , offset δ , edge radius r , edge mode m
Result: watertight triangle mesh T

```
1  $C \leftarrow \text{largest\_contour}(M)$ ;  
2  $\{\mathbf{p}_i\}_{i=1}^N \leftarrow \text{spline}(C)$  ; /* closed degree-5 B-spline,  $N = 300$  */  
3  $\mathbf{p}_i \leftarrow \mathbf{p}_i/s$ ,  $i = 1, \dots, N$  ; /*  $\text{px} \rightarrow \text{mm}$  */  
4  $\bar{\mathbf{c}} \leftarrow \frac{1}{N} \sum_{i=1}^N \mathbf{p}_i$ ;  
5 for  $i \leftarrow 1$  to  $N$  do  
6    $\hat{\mathbf{u}}_i \leftarrow \frac{\mathbf{p}_i - \bar{\mathbf{c}}}{\|\mathbf{p}_i - \bar{\mathbf{c}}\|}$ ;  
7    $\mathbf{p}_i \leftarrow \mathbf{p}_i + \delta \hat{\mathbf{u}}_i$  ; /* perilesional offset */  
8  $R^\uparrow \leftarrow \{(\mathbf{p}_i, t)\}$ ,  $R^\downarrow \leftarrow \{(\mathbf{p}_i, 0)\}$ ;  
9 if  $r > 0$  then  
10 |  $\mathcal{R} \leftarrow \text{edge\_rings}(\{\mathbf{p}_i\}, r, m)$ ;  
11 else  
12 |  $\mathcal{R} \leftarrow \emptyset$ ;  
13  $T \leftarrow \text{triangulate}(R^\uparrow, \mathcal{R}, R^\downarrow)$ ;  
14 return  $T$ ;
```

3.4. Segmentation back-ends and reference

The automatic stage offers three interchangeable back-ends, a promptable foundation model and two classical region methods, so the geometric pipeline can be exercised independently of any single segmenter. For a fair comparison the three are seeded from the same manual reference, isolating the boundary-refinement behaviour of each under a common initialization rather than its ability to localize the wound unaided. When no manual delineation is available, a reference-free prompt is derived automatically by Otsu thresholding [22], a border-based polarity check and selection of the largest interior connected component while discarding border-touching blobs, so the tool also runs without an operator.

The reference is delineated by an operator who places a variable number of control points, depending on the wound complexity, interpolated by a closed periodic cubic interpolating spline resampled to $N = 300$ points, then rasterized by polygon filling. This is the same sample count and resampling as the automatic contour, so manual and automatic masks pass through the identical operator of Algorithm 1. The delineation was performed by a single operator, a point revisited in Section 5.

The first back-end is the Segment Anything Model (SAM, ViT-B) [30], prompted with the axis-aligned bounding box of the reference contour and a single positive point placed at the maximum of the distance transform rather

than at the centroid, since for concave contours the centroid can fall outside the mask whereas the distance-transform maximum is interior by construction; multiple-mask output is disabled. The second is GrabCut [24], initialized with probable foreground inside the reference region, certain foreground on a core eroded by two iterations and certain background outside a band dilated by two iterations, run for five iterations and restricted to the dilated band. The third is the marker-controlled watershed of Meyer [23], with the eroded core as foreground markers and the region outside a dilated band as background, the intervening band resolved by flooding. In all cases the morphological radius is tied to the image size, $k = \max(7, 0.02 \max(H, W))$ px, and the raw output is cleaned by an opening and closing with an elliptical kernel $k = \max(3, 0.012 \max(H, W))$ px forced odd, followed by retention of the largest connected component, before entering the geometric chain.

3.5. Evaluation metrics

The manual and automatic results are compared at two levels, mask agreement and clinical coverage. Areas and distances are expressed in millimetres through s where a physical reference is available; the clinical images carry no such reference, so s is undefined there and the clinical metrics (Table 2, including the Hausdorff distance) are reported in pixels, while millimetre quantities are reported only for the calibrated phantom experiments. Let G and S be the reference and automatic masks, with boundaries $\partial G, \partial S$, and $|\cdot|$ the pixel count. Region overlap is summarized by the Dice coefficient [32] and the intersection over union,

$$\text{Dice}(G, S) = \frac{2|G \cap S|}{|G| + |S|}, \quad (3)$$

$$\text{IoU}(G, S) = \frac{|G \cap S|}{|G \cup S|}. \quad (4)$$

Size agreement is the relative area error on the calibrated areas,

$$A_X = \frac{|X|}{s^2}, \quad (5)$$

$$\varepsilon_A = \frac{|A_G - A_S|}{A_G}. \quad (6)$$

The worst-case boundary deviation is the symmetric Hausdorff distance [33], built from the directed distance $\vec{h}$,

$$\vec{h}(X, Y) = \max_{x \in X} \min_{y \in Y} \|x - y\|, \quad (7)$$

$$H(\partial G, \partial S) = \frac{1}{s} \max\{\vec{h}(\partial G, \partial S), \vec{h}(\partial S, \partial G)\}. \quad (8)$$

Finally, because the dressing must cover the wound, the disagreement is split into the clinically unsafe under-coverage A_u (wound left exposed) and the conservative over-coverage A_o (excess on healthy skin), with coverage fraction c ,

$$A_u = \frac{|G \setminus S|}{s^2}, \quad (9)$$

$$A_o = \frac{|S \setminus G|}{s^2}, \quad (10)$$

$$c = 1 - \frac{A_u}{A_G} = \frac{|G \cap S|}{|G|}, \quad (11)$$

as illustrated in Fig. 4(d). Overlap alone is an incomplete predictor of geometric fidelity, since two masks with equal Dice can differ markedly in boundary deviation, and hence in the fabricated patch (Fig. 4(b,c)); overlap and boundary measures are therefore reported jointly and their relationship to the patch error is examined in Section 4.4. The released tool also reports shape descriptors and further boundary measures (perimeter error and average symmetric surface distance), omitted here for compactness.

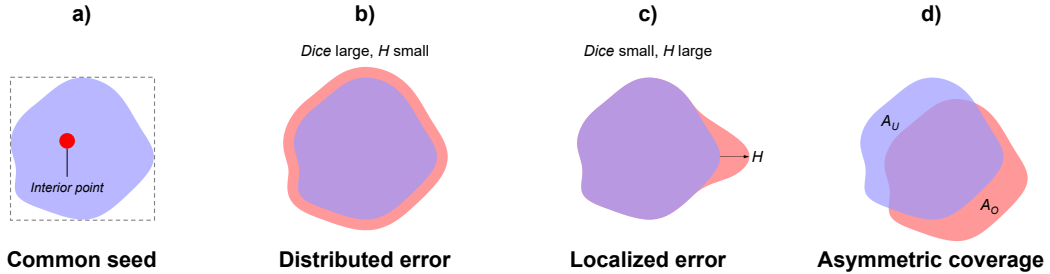


Figure 4: Concept of the fabrication-aware evaluation: (a) the back-ends share the operator seed; (b,c) two masks with the same Dice differ in boundary deviation H ; (d) the coverage split into under-coverage A_u and over-coverage A_o .

3.6. Error budget and offset sizing

The fabricated-patch error is decomposed into three terms, each measured against a known reference at its stage: a segmentation term E_{Seg} (relative patch-area error of the automatic mask against the reference at fixed scale), a scale term E_{Scale} (relative area error from calibration-point placement, Section 4.3) and a printing term E_{Pr} (relative in-plane area error of the part against its STL, Section 4.2). The segmentation term is computed on the outlier-removed clinical set of Table 2, so that the budget and the per-back-end accuracy are reported on the same data. Treated as independent, they combine by a root-sum-of-squares budget,

$$E_{\text{patch}} = \sqrt{E_{\text{Seg}}^2 + E_{\text{Scale}}^2 + E_{\text{Pr}}^2}. \quad (12)$$

The three terms are obtained from separate, dedicated experiments, each against its own reference, rather than from a single wound carried through the whole pipeline. A real wound has no physical ground-truth patch, so the manual reference is taken as the proxy ground truth and E_{Seg} quantifies agreement with that delineation rather than with the true wound boundary. In the per-image patch-area error the scale cancels, because the reference and automatic patches are built with the same s (Algorithm 2); E_{Seg} therefore measures the automatic-versus-reference shape discrepancy at a fixed, assumed scale, E_{Scale} the absolute-scale error of the calibration, and E_{Pr} the part-versus-STL printing error. Their root-sum-of-squares is thus a first-order, independence-assuming estimate of the fabricated-patch error rather than a single end-to-end measurement, and it omits terms not exercised here, notably the operator-versus-true-boundary error and any departure from planarity (Section 5). Table 4 reports the representative (mean) magnitudes combined in quadrature, whereas Fig. 8 decomposes the corresponding variances; Eq. (12) is therefore a representative combination of magnitudes under independence, while the variance propagation proper, in the sense of the guide to uncertainty in measurement [14], is the decomposition of Fig. 8. The complete data flow is shown in Fig. 5 and the evaluation loop over images and back-ends in Algorithm 2, in which METRICS returns Dice, IoU, ε_A , H , A_u and A_o (Eqs. (3)–(11)), $a(\cdot)$ is the in-plane area and CHAIN is Algorithm 1. Because under-coverage is the unsafe error, the offset δ is sized from its empirical distribution rather than fixed arbitrarily: for each image i , ρ_i is the maximum inward deviation of the automatic contour with respect to the reference, that is the smallest outward margin that fully covers the wound, and the offset

achieving coverage in a fraction τ of cases is the empirical τ -quantile of these margins,

$$\delta(\tau) = Q_\tau(\{\rho_i\}), \quad (13)$$

evaluated per back-end, with $\tau = 0.95$.

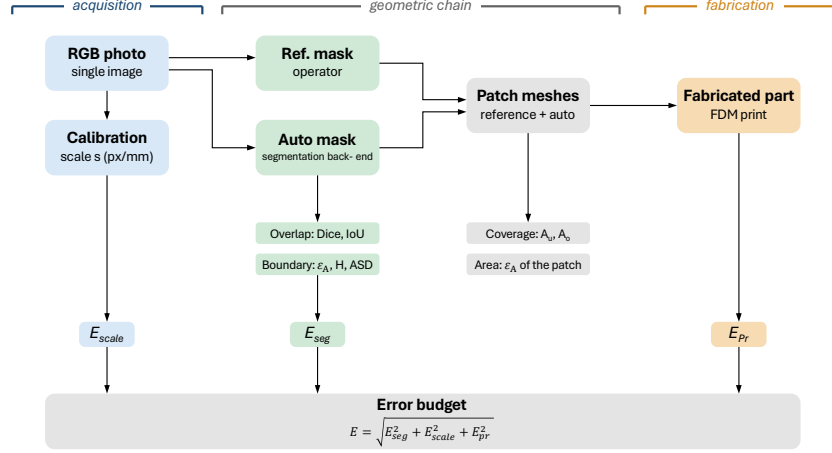


Figure 5: Data flow of the stage-wise error budget: the three terms E_{Seg} , E_{Scale} and E_{Pr} are measured on separate experiments and combined into the fabricated-patch error (Eq. 12).

3.7. Experimental design

The pipeline was evaluated in three experiments and a calibration-sensitivity analysis, each isolating one stage of the budget.

Experiment A (clinical segmentation and propagation). A set of 149 clinical DFU photographs was assembled from two public sources, the Lower Limb and Feet Wound Image Dataset [36] and a public diabetic foot ulcer image dataset [37], retaining superficial, single and clearly delineable wounds acquired in an approximately fronto-parallel view. Each image was manually delineated by a single operator to give the reference, then segmented by the three back-ends; the mask-level metrics and the propagated patch-level area error were computed (RQ1, RQ2), with the perilesional offset set to $\delta = 0$ so that the offset, derived in Section 3.6 and applied in RQ3, does not enter the segmentation-error characterization.

Experiment B (absolute geometric accuracy). Six planar shapes of known CAD geometry, each printed in three replicates (18 parts), were photographed under the same calibrated protocol, and the reconstructed

Algorithm 2: Dataset evaluation across segmentation back-ends

Data: image set $\mathcal{I}$, reference length d , back-end set $\mathcal{B}$, parameters (t, δ, r, m)

Result: per-image, per-back-end record set $\mathcal{R}$

```
1  $\mathcal{R} \leftarrow \emptyset$ ;
2 for each image  $I \in \mathcal{I}$  do
3    $s \leftarrow \text{calibrate}(I, d)$ ; /* two-point scale, Eq. (1) */
4    $M_G \leftarrow \text{manual\_reference}(I)$ ;
5    $T_G \leftarrow \text{chain}(M_G, s, t, \delta, r, m)$ ; /* reference patch, Alg. 1 */
6   for each back-end  $B \in \mathcal{B}$  do
7      $M_S \leftarrow \text{auto\_seg}(I, M_G, B)$ ; /* seeded from  $M_G$  */
8     if  $M_S = \emptyset$  then  $\mathcal{R} \leftarrow \mathcal{R} \cup \{(I, B, \text{fail})\}$ ; continue;
9      $\mathcal{M} \leftarrow \text{metrics}(M_G, M_S, s)$ ; /* Eqs. (3)-(11) */
10     $T_S \leftarrow \text{chain}(M_S, s, t, \delta, r, m)$ ;
11     $\varepsilon_{\text{patch}} \leftarrow \frac{|a(T_S) - a(T_G)|}{a(T_G)}$ ; /* fabricated-patch area error */
12     $\mathcal{R} \leftarrow \mathcal{R} \cup \{(I, B, \mathcal{M}, \varepsilon_{\text{patch}})\}$ ;
13 return  $\mathcal{R}$ ;
```

in-plane geometry was regressed against the nominal CAD size. This isolates the joint segmentation-and-calibration error on objects of known geometry, before printing.

Experiment C (printing accuracy). The same 18 parts were measured against their source STL files to isolate E_{Pr} , separating the in-plane area error from the out-of-plane thickness, which is a user-set parameter rather than a reconstructed quantity.

Scale-calibration sensitivity. A 500-sample Monte-Carlo perturbation displaced the two calibration points by Gaussian noise of standard deviation σ and recorded the relative area error, characterizing E_{Scale} ; the representative value $\sigma = 3$ px is adopted in the budget. The relative scale error grows in proportion to σ/d_{px} , with d_{px} the reference length in pixels, so σ is reported in absolute pixels and the resulting scale term reflects the reference lengths present in the dataset rather than a single fixed physical reference.

Phantoms were fabricated by fused-deposition modelling in PLA on a Creality K1C printer at 0.2 mm layer height; planar reference dimensions were measured with a digital caliper. All clinical references were delineated by a single operator, a limitation discussed in Section 5.

3.8. Statistical analysis

Normality was assessed with the Shapiro–Wilk test; the per-image scores departed from normality (Shapiro–Wilk, $p < 0.05$), so non-parametric tests were used throughout. Per-image outliers were removed before Table 2 by

Tukey’s rule, discarding values beyond 1.5 IQR from the quartiles of the per-image patch-area error; the same cases are excluded from the overlap and boundary columns so that all metrics of Table 2 describe one image set. Differences across the three back-ends are evaluated with the Friedman omnibus test on the per-image scores of the cases segmented without outliers by all three back-ends, followed by post-hoc Wilcoxon signed-rank tests with Bonferroni correction for the three pairwise comparisons. The relationship between mask overlap and the fabricated-patch error (RQ2) is summarized by the Spearman rank correlation and the coefficient of determination of a linear fit. Absolute geometric accuracy on the phantoms is reported by linear regression of reconstructed against nominal size. The significance level is $\alpha = 0.05$, and the analyses use Python with standard scientific libraries. The reference is delineated by a single operator, so no inter-operator agreement is reported (Section 5).

4. Results

Findings are reported in the order of the three research questions; their interpretation is deferred to Section 5. The per-image overlap and patch-area scores of Section 4.1 are the inputs to the error budget (RQ1, Section 4.3) and to the overlap-versus-error analysis (RQ2, Section 4.4).

4.1. Segmentation accuracy across back-ends

Watershed failed on three images (`case_006`, `case_038`, `case_039`); SAM and GrabCut returned a mask on all 149. Per-image scores (per-back-end outliers removed as defined in Section 3.8: 5, 14 and 10 cases for SAM, GrabCut and watershed) are summarized in Table 2 and Fig. 6. SAM and GrabCut did not differ in Dice or IoU (Bonferroni-adjusted paired Wilcoxon, $p > 0.05$); GrabCut exceeded Watershed (Dice and IoU, $p < 10^{-4}$), and SAM and Watershed did not differ. For the patch area error GrabCut was lower than both SAM and Watershed ($p < 10^{-3}$), which did not differ from each other; for the Hausdorff distance only the GrabCut–Watershed pair reached significance ($p < 10^{-2}$). All Friedman omnibus tests were significant ($p < 10^{-3}$; for Dice, $\chi^2 = 24.98$). The patch area error and the Hausdorff distance were right-skewed (for SAM, medians 5.2% and 4.6 px against means of 6.4% and 8.2 px).

Table 2: Segmentation accuracy on the 149 clinical DFU images (mean $\pm$ SD, outliers removed).

Back-end	n	Dice	IoU	Patch area error (%)	Hausdorff (px)
SAM	144	0.950 ± 0.025	0.907 ± 0.044	6.39 ± 4.50	8.18 ± 10.41
GrabCut	135	0.953 ± 0.024	0.912 ± 0.042	5.01 ± 4.69	7.65 ± 8.09
Watershed	136	0.942 ± 0.029	0.892 ± 0.051	6.79 ± 5.29	9.14 ± 8.59

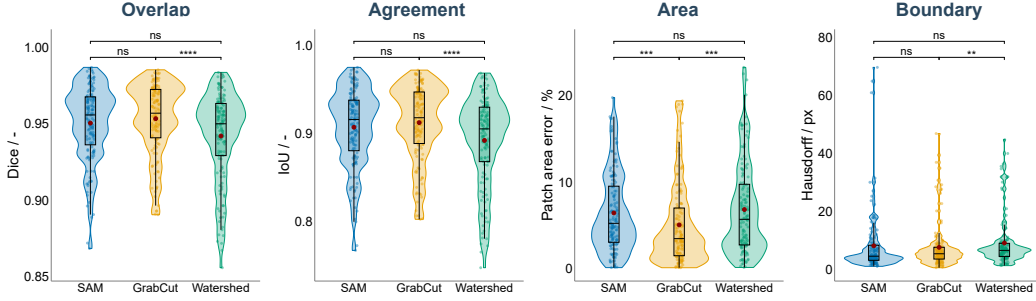


Figure 6: Per-image Dice, IoU, patch area error and Hausdorff distance per back-end ($n = 144, 135, 136$). Red dots are means; brackets give Bonferroni-adjusted paired Wilcoxon results (ns; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$).

4.2. Absolute geometric and printing accuracy

On the 18 phantoms (six planar CAD shapes printed in three replicates, Fig. 7) the reconstructed in-plane size regressed against the nominal CAD with slope 0.994 and $R^2 = 0.9995$ (Table 3): mean dimensional error 0.80%, area error 0.32%, Dice 0.997, with a worst-case minor-axis underestimate of 2.51% on the crescent. Measured against the source STL, the parts had a mean in-plane area error of 0.69% and a dimensional error of $0.41 \pm 0.11\%$; the out-of-plane thickness deviated by +7.4% and is reported apart from the in-plane budget.

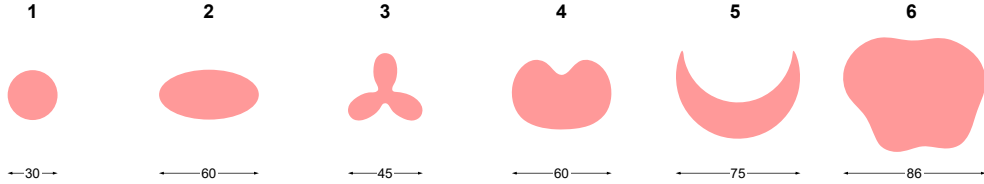


Figure 7: Nominal CAD geometries of the six phantom shapes (disc, ellipse, trefoil, bean, crescent and freeform), with the nominal in-plane size in millimetres.

Table 3: Absolute geometric accuracy on the 18 phantoms (Exp. B) and printing accuracy against the STL (Exp. C).

Quantity	Value
Regression slope (reconstructed vs nominal)	0.994
R^2	0.9995
Mean dimensional error (Exp. B)	0.80%
Mean area error (Exp. B)	0.32%
Mean Dice (Exp. B)	0.997
Worst-case minor-axis underestimate (crescent)	2.51%
Printing in-plane area error (Exp. C)	0.69%
Printing dimensional error (Exp. C)	$0.41 \pm 0.11\%$

4.3. Dominant error source (RQ1)

The stage magnitudes and their root-sum-of-squares combination are in Table 4: the segmentation term was 6.39%, 5.01% and 6.79% (the outlier-removed means of Table 2), the scale term 5.0% at $\sigma = 3$ px, the printing term 0.69%, and the combined error 8.14%, 7.11% and 8.46%. The scale-calibration sensitivity grew about twice as fast as the scale relative error (Fig. 8, right). In the first-order variance decomposition (Fig. 8, left) the printing term was negligible ($< 0.1\%$); the scale term was the larger contributor for SAM (59%) and GrabCut (53%) and the segmentation term for Watershed (52%). The total fabricated-patch area error was right-skewed, with a median near 7% (Fig. 9, left), exceeding 5% in roughly two thirds of cases and 10% in roughly a third (Fig. 9, right).

Table 4: Stage error magnitudes (relative area error) and their root-sum-of-squares combination.

Back-end	E_{Seg} (%)	E_{Scale} (%)	E_{Pr} (%)	Combined (%)
SAM	6.39	5.0	0.69	8.14
GrabCut	5.01	5.0	0.69	7.11
Watershed	6.79	5.0	0.69	8.46

4.4. Overlap as a predictor of fabricated-patch error (RQ2)

Pooling all valid cases ($n = 444$, without the per-back-end outlier removal applied in Table 2, retained for this analysis to span the full range of behaviour including the high-error cases on which the residual-spread argument below

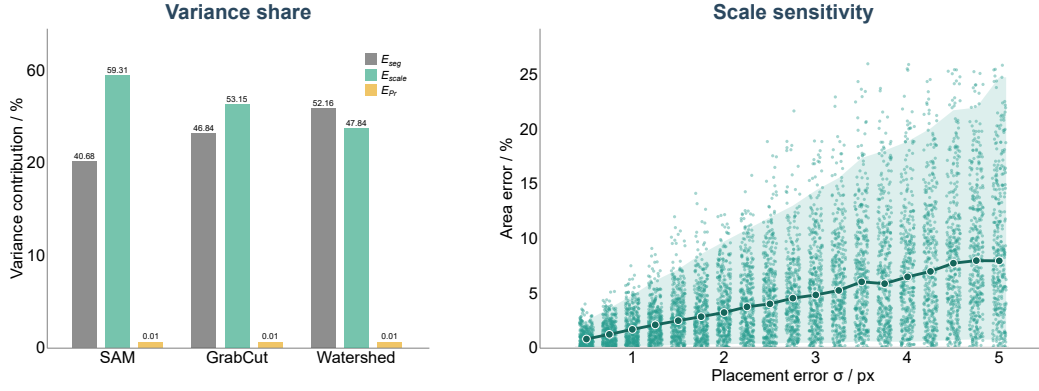


Figure 8: Left, first-order variance contribution of the segmentation, scale and printing terms to the fabricated-patch area error. Right, scale-calibration sensitivity (area error against placement uncertainty σ).

relies), mask overlap was correlated with, but did not determine, the fabricated-patch area error. The Dice coefficient and the patch area error were negatively and monotonically associated (Spearman $\rho = -0.68$, $p < 10^{-3}$; an ordinary-least-squares fit gave $R^2 = 0.76$), and the association held within each back-end ($\rho = -0.62$, -0.65 and -0.75 for SAM, GrabCut and Watershed; Fig. 10). At a fixed overlap the patch error still varied over a wide range: among the 235 cases with Dice ≥ 0.95 the patch area error spanned 0 to 8.9% (median 3.3%, IQR 1.7–5.2%). Consistently, the two back-ends with statistically indistinguishable Dice (SAM and GrabCut, Section 4.1) differed significantly in patch area error ($p < 10^{-3}$), and the back-end ordering by overlap (GrabCut and SAM highest, Watershed lowest) did not match the ordering by patch area error (GrabCut lowest, SAM and Watershed not differing).

4.5. Coverage and perilesional offset (RQ3)

At zero offset the mean areal coverage was 92.6%, 95.0% and 92.1% for SAM, GrabCut and Watershed, and almost no case was already fully covered (at most 2% of cases above 99.5% coverage), so an outward margin is needed in essentially every case. The disagreement was clinically asymmetric (Fig. 11): under-coverage (exposed wound) was 7.4%, 5.0% and 7.9% and over-coverage (excess healthy skin) 2.2%, 4.3% and 3.4%; under-coverage exceeded over-coverage for SAM and Watershed (paired Wilcoxon, $p < 10^{-4}$) but not for GrabCut ($p = 0.07$). From the per-image inward deviations of

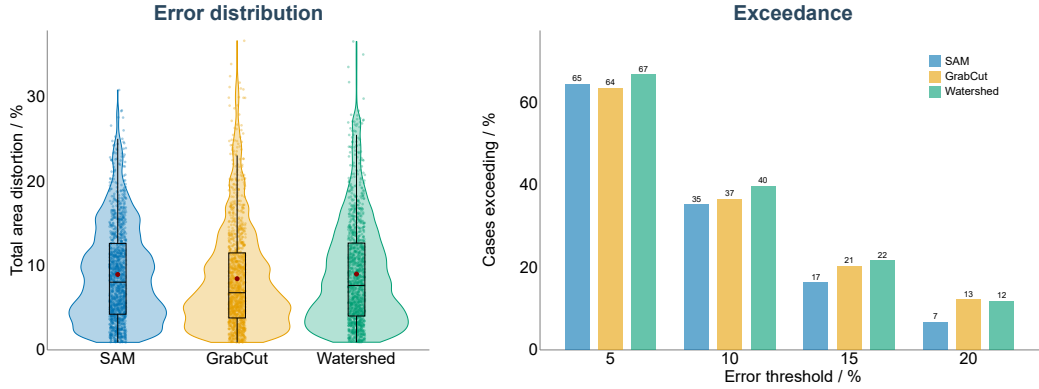


Figure 9: Left, distribution of the total fabricated-patch area error per back-end. Right, probability that the total error exceeds a given threshold. Because the clinical images are uncalibrated and carry no per-image scale error, the per-image total combines each image’s clinical segmentation error in quadrature with the representative scale term (5%, from the phantom-based Monte-Carlo calibration) and the printing term (0.69%); its empirical per-image median therefore lies below the combined value of Table 4, which combines the mean magnitudes.

the automatic contour with respect to the reference, the perilesional offset of Eq. (13) covering the wound in 95% of cases was taken as the 95th percentile of those deviations, evaluated separately for each back-end.

5. Discussion

IMTOP turns a single calibrated RGB photograph into a watertight, printable patient-specific dressing without depth sensing, and is evaluated on the fabricated object rather than on the segmentation mask alone. For the phantoms this object is a physically measured part; for the clinical images, where no physical patch can be printed from a real wound for want of a ground-truth object, it is the patch-area error propagated through the chain, the phantoms having shown the printing contribution to be negligible. On 149 clinical DFU images and 18 printed phantoms the pipeline reconstructed the planar geometry accurately (phantom area error 0.32%, Dice 0.997), and the combined fabricated-patch area error was 8.1% for the default back-end. The three research questions, whose results were reported without interpretation in Section 4, are addressed in turn below.

RQ1 asked how the fabricated-patch error distributes across the segmentation, scale and printing stages, in magnitude and in variance.

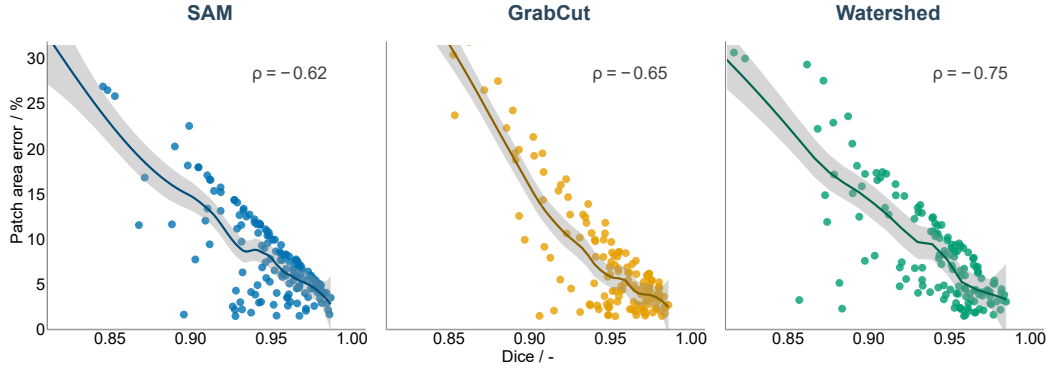


Figure 10: Fabricated-patch area error against Dice per back-end ($n = 444$); the line is the running median. The few points above 30% are clipped from the view only.

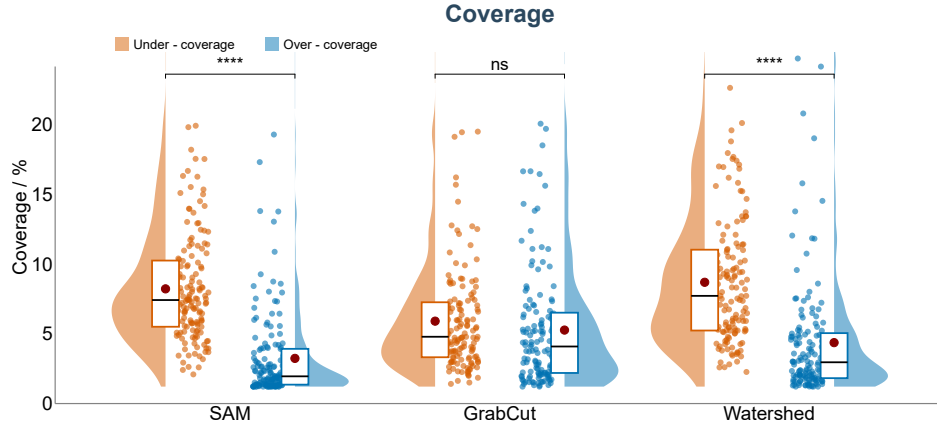


Figure 11: Under-coverage (exposed wound) and over-coverage (excess on healthy skin) at zero offset, per back-end. Red dots are means.

Printing is negligible by both magnitude (0.69%) and variance ($< 0.1\%$), so the budget is governed by segmentation and scale. By typical magnitude the segmentation term is the largest (6.4% against 5.0% for scale), whereas in the variance decomposition the scale term, despite its smaller median, dominates for SAM and GrabCut because of its right-skewed upper tail, segmentation dominating only for watershed. This variance ranking is contingent on the assumed calibration-placement noise $\sigma = 3$ px, which is expressed in absolute pixels and is therefore resolution-dependent; at a smaller σ the scale variance would shrink and segmentation would dominate, so the ranking should be read as conditional on σ rather than absolute. That the segmentation term is large on clinical images yet small on phantoms with sharp borders (0.32%) locates

it in the ambiguity of the wound boundary itself, not in the segmentation algorithm or the scale recovery. Stage-wise error propagation of this kind follows the uncertainty-propagation guide [14] and the dimensional-tolerance studies of fused-deposition modelling by Lienenke et al. [15] and Abas et al. [34], and we previously applied it to surgical guides [35]; to our knowledge it has not been applied to image-to-print wound dressings.

RQ2 asked whether mask overlap suffices to predict the fabricated-patch error. It does not. Dice was correlated with but did not determine the patch area error (Spearman $\rho = -0.68$), and at high overlap the patch error remained dispersed, reaching 8.9% among the cases with Dice ≥ 0.95 ; the two back-ends with statistically indistinguishable Dice differed significantly in patch area error, and the overlap and patch-error rankings disagreed. Region overlap is therefore necessary but insufficient, and boundary and coverage measures are needed to characterize the fabricated object. This refines the common practice of ranking wound and DFU segmentation by overlap alone, from Goyal et al. [11] and Wang et al. [10] to the DFU segmentation challenge [29] and self-configuring nnU-Net [28], and is consistent with the insensitivity of Dice to boundary placement noted by Huttenlocher et al. [33].

RQ3 asked how the residual segmentation uncertainty should set the perilesional margin. Because the disagreement is clinically asymmetric, under-coverage of the wound being the unsafe error and exceeding over-coverage for SAM and watershed, the offset is sized from the empirical distribution of inward deviations as their per-back-end 95th percentile, covering the wound in 95% of the evaluated cases. With a mean coverage of 92 to 95% at zero offset and almost no case already fully covered, such a margin is needed in essentially every case. This percentile is computed in-sample, on the same 149 images and a single-operator reference and without fixing its value in millimetres, so it is a constructive sizing rule rather than a guarantee, and an out-of-sample, calibrated validation is left to future work. It converts segmentation uncertainty into a constructive design rule, complementary to the assessment and monitoring step that precedes fabrication [21].

Set against the existing routes, IMTOP is a deliberate trade-off rather than a strict improvement. Image-based planimetry recovers wound geometry from a single photograph but stops at a measurement: Foltynski reports area [13], Yap et al. standardize smartphone acquisition [16], and Kim et al. predict healing [18], none producing a fabricable object, whereas IMTOP carries the same depth-free image through to a printable patch. At the other end, Chae et al. [9] also reach a printed DFU patch, but from a reconstructed

RGB-D surface: their depth capture recovers the true wound-bed relief that our planar assumption gives up, while our single-photograph input removes their depth sensor. Filko et al. [19, 20] likewise reconstruct and measure the wound from a robot-mounted RGB-D rig, validating on the reconstructed surface rather than, as here, on the fabricated part. Cavazzana et al. [21], from the same group, segment and analyse the ulcer in three dimensions for objective assessment; IMTOP is complementary to this, the fabrication step that can follow such an assessment rather than an alternative to it. Finally, the three back-ends are seeded from the same manual reference and therefore refine a common initialization rather than localizing the wound independently; their close agreement in overlap, with SAM [30] and GrabCut [24] differing little, is thus partly a consequence of the shared seed, and the observation that the choice of segmenter matters less than the wound-border ambiguity common to all of them is correspondingly qualified and would require a reference-free comparison to be established.

Several limitations bound these results. First, the reference was delineated by a single operator with a geometric rather than clinical background; because E_{Seg} is measured against this single reference, it cannot separate the intrinsic wound-border ambiguity from the idiosyncrasy of one tracer, so the attribution of the dominant error to border ambiguity is provisional until multi-annotator, clinician-validated references are available. Second, the error budget is partial: it omits the operator-versus-true-boundary term just noted and any departure from planarity, since the pipeline assumes an approximately fronto-parallel view and the phantoms are planar, so the 0.32% phantom area error is a planar best case and does not bound the error on a curved foot. Third, the clinical images were uncalibrated, so the clinical metrics are in pixels and the offset is reported as a method whose absolute millimetre value awaits a calibrated sizing run; the three error terms are also treated as independent. Fourth, the printed thickness deviated by +7.4%: although it is a user-set parameter excluded from the in-plane budget, for a hydrogel the dispensed volume scales with thickness, so this deviation is functionally relevant to local drug dosing. Finally, no established clinical threshold for acceptable patch-area error exists, so the reported combined error of about 8% cannot yet be declared adequate or inadequate for treatment. Future work will add multi-annotator and clinician-validated references, characterize the calibration-placement repeatability to ground the value of σ , fix the offset in millimetres on calibrated images and validate it prospectively, bound the planarity error on curved or obliquely viewed targets, optionally incorporate depth where

available without sacrificing the frugal default, and integrate the patch with the glucose-responsive hydrogel ink toward a printed dressing tested on the wound.

6. Conclusions

A single calibrated RGB photograph is sufficient to produce a patient-specific DFU dressing, validated on the fabricated object and without depth sensing. The dominant geometric error arises from the ambiguity of the wound boundary rather than from scale recovery or printing, mask overlap alone does not fully determine fabricated-patch accuracy, and the residual segmentation uncertainty yields a constructive perilesional offset that covers the wound at a target level on the evaluated data. The application will be released to support reproducibility (Code availability), and ongoing work targets clinician-validated delineation and integration with a glucose-responsive printable hydrogel.

Declaration of interests

The authors declare no conflicts of interest.

Code availability

The IMTOP application and the analysis scripts will be made publicly available on GitHub ([public link](#)) and archived on Zenodo with a permanent DOI upon publication. The SAM ViT-B checkpoint is distributed by its original authors [30] and is not redistributed here.

Data availability

The clinical images are drawn from two public datasets, the Lower Limb and Feet Wound Image Dataset [36] (DOI 10.17632/hsj38fwnvr.3) and a public diabetic foot ulcer image dataset [37], and are not redistributed here. The phantom CAD models, STL files and dimensional measurements will be released together with the application (Code availability).

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